

Ex 2 — Working with Pandas

Dataset: fanspeed.csv
Reg No: URK24CS1021

```
print("URK24CS1021")  
  
import pandas as pd  
  
df = pd.read_csv("fanspeed.csv")  
df
```

URK24CS1021

	Temperature	Observed Fan Speed
0	5.000000	22.483571
1	5.303030	19.914739
2	5.606061	24.450564
3	5.909091	29.433331
4	6.212121	21.253476
...	...	...
95	33.787879	104.046062
96	34.090909	113.753329
97	34.393939	114.487094
98	34.696970	114.116476
99	35.000000	113.827064

100 rows × 2 columns

```
print("URK24CS1021")  
  
df.head(10)
```

URK24CS1021

	Temperature	Observed Fan Speed
0	5.000000	22.483571
1	5.303030	19.914739
2	5.606061	24.450564
3	5.909091	29.433331
4	6.212121	21.253476
5	6.515152	21.859618
6	6.818182	31.532428
7	7.121212	28.079598
8	7.424242	22.501113
9	7.727273	28.167346

```
print("URK24CS1021")  
  
df.tail(8)
```

URK24CS1021

	Temperature	Observed Fan Speed
92	32.878788	105.126098
93	33.181818	107.907144
94	33.484848	108.494005
95	33.787879	104.046062
96	34.090909	113.753329
97	34.393939	114.487094
98	34.696970	114.116476
99	35.000000	113.827064

```
print("URK24CS1021")

df.shape, list(df.columns), df.dtypes
```

URK24CS1021
((100, 2), ['Temperature', 'Observed Fan Speed'], Temperature float64
Observed Fan Speed float64
dtype: object)

```
print("URK24CS1021")

df.describe()
```

URK24CS1021

	Temperature	Observed Fan Speed
count	100.000000	100.000000
mean	20.000000	66.568646
std	8.791361	30.065419
min	5.000000	18.312387
25%	12.500000	39.894309
50%	20.000000	69.669067
75%	27.500000	93.760259
max	35.000000	114.487094

```
print("URK24CS1021")

demo = df.copy()
demo.loc[[3, 27, 61], "Observed Fan Speed"] = None
demo.isnull().sum()
```

URK24CS1021
Temperature 0
Observed Fan Speed 3
dtype: int64

```
print("URK24CS1021")

demo["Observed Fan Speed"] = demo["Observed Fan Speed"].fillna(demo["Observed Fan Speed"].mean())
demo.isnull().sum()
```

URK24CS1021
Temperature 0
Observed Fan Speed 0
dtype: int64

```
print("URK24CS1021")

df.loc[5:8, ["Temperature"]]
```

URK24CS1021

	Temperature
5	6.515152
6	6.818182
7	7.121212
8	7.424242

```
print("URK24CS1021")

df[df["Temperature"] > 25].head()
```

URK24CS1021

	Temperature	Observed Fan Speed
67	25.303030	90.926755
68	25.606061	88.626362
69	25.909091	84.501674
70	26.212121	90.443342
71	26.515152	97.235637

```
print("URK24CS1021")

df["Speed_Per_Degree"] = df["Observed Fan Speed"] / df["Temperature"]
df["Speed_Level"] = df["Observed Fan Speed"].apply(lambda x: "High" if x > 60 else "Low")
df.head()
```

URK24CS1021

	Temperature	Observed Fan Speed	Speed_Per_Degree	Speed_Level
0	5.000000	22.483571	4.496714	Low
1	5.303030	19.914739	3.755351	Low
2	5.606061	24.450564	4.361452	Low
3	5.909091	29.433331	4.981025	Low
4	6.212121	21.253476	3.421291	Low

```
print("URK24CS1021")

df.sort_values("Observed Fan Speed", ascending=False).head()
```

URK24CS1021

	Temperature	Observed Fan Speed	Speed_Per_Degree	Speed_Level
97	34.393939	114.487094	3.328700	High
98	34.696970	114.116476	3.288946	High
99	35.000000	113.827064	3.252202	High
96	34.090909	113.753329	3.336764	High
91	32.575758	112.570498	3.455652	High

```
print("URK24CS1021")

df.groupby("Speed_Level")["Observed Fan Speed"].mean()
```

URK24CS1021
Speed_Level
High 88.332559
Low 35.249845
Name: Observed Fan Speed, dtype: float64

```
print("URK24CS1021")

df["Rank_dense"] = df["Observed Fan Speed"].rank(method="dense")
df.sort_values("Rank_dense").head()
```

URK24CS1021

	Temperature	Observed Fan Speed	Speed_Per_Degree	Speed_Level	Rank_dense
13	8.939394	18.312387	2.048504	Low	1.0
14	9.242424	19.860259	2.148815	Low	2.0
1	5.303030	19.914739	3.755351	Low	3.0
.	0.000000	0.000000	0.000000	.	0.0